BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

BIOLOGY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Biology (Objective) Group - I

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8465

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. In DNA structure are more strongly linked:

(A) Adenine with thymine
(B) Guanine with cytosine
(C) Bases with sugar
(D) All these

Q2. Intercalary meristems are situated at:

(A) Root apex
(B) Shoot apex
(C) Top of internode
(D) Base of internode

Q3. Germ layers are formed at:

(A) Cleavage
(B) Gastrulation
(C) Organogenesis
(D) Neurulation

Q4. The plants which are stimulated to flower by exposure to low temperature are said to have:

(A) Vernalization
(B) Parthenocarpy
(C) Parthenogenesis
(D) Apomixis

Q5. Adrenocorticotrophic hormone (ACTH) is secreted by:

(A) Adrenal gland
(B) Hypothalamus
(C) Pituitary gland
(D) Thyroid gland

Q6. Bone dissolving cells are:

(A) Osteoblasts
(B) Osteoclasts
(C) Osteocytes
(D) Chondrocytes

Q7. Disease in which bone resorption outpaces bone deposit:

(A) Osteoporosis
(B) Osteomalacia
(C) Rickets
(D) Spondylosis

Q8. Liver functions are pivotal to:

(A) Osmo-regulation
(B) Excretion
(C) Homeostasis
(D) Thermo-regulation

Q9. As CFCs rise to atmosphere, ultraviolet rays release:

(A) Chlorine
(B) Fluorine
(C) Carbon
(D) Hydrogen

Q10. The desert of Southern Punjab is:

(A) Thal
(B) Sahara
(C) Thar
(D) Cholistan

Q11. The conversion of nitrate to ammonium with in plant cell is called:

(A) Ammonification
(B) Nitrification
(C) Assimilation
(D) Denitrification

Q12. Prokaryotes have arisen more than:

(A) 1.5 billion years ago
(B) 2.5 billion years ago
(C) 3.5 billion years ago
(D) 4.5 billion years ago

Q13. If correct proportions of auxin and cytokinin are added in a liquid medium, thousands of copies of new shoots will develop from a single shoot tip by:

(A) Protoplast culture technique
(B) Meristem culture technique
(C) Anther culture technique
(D) Cell suspension technique

Q14. Which one of them is mostly used to develop transgenic animal?

(A) Sanger method
(B) Maxam Gilbert method
(C) Particle gun method
(D) Micro injection method

Q15. What is the risk of a haemophiliac child in a family when father is haemophiliac but mother is carrier?

(A) All sons normal
(B) All sons affected
(C) All sons affected but all daughters normal
(D) Half sons and half daughters affected

Q16. The actual decrease of chromosome number occur in:

(A) Meiosis-I
(B) Meiosis-II
(C) Mitosis
(D) Cytokinesis

Q17. Which tumor has branches:

(A) Malignant
(B) Benign
(C) Both
(D) None of these

Biology (Subjective) Group - I

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 8465

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) Distinguish between osmoconformers and osmoregulators.

(ii) Compare the nitrogen excreted in lower quantities with the one excreted in very small quantities.

(iii) Differentiate between protonephridium and metanephridium. Give examples.

(iv) What is vascular cambium? Which new tissues develop from it?

(v) What is phototactic movement? Give example.

(vi) Define vertebral column. What names can be given to them according to their location?

(vii) Differentiate between identical and fraternal twins.

(viii) Which hormone is released by pituitary gland at puberty? Also define follicle atresia.

(ix) Write a note on profundal zone. Which organisms inhabit this zone?

(x) Write the names of any eight animals which inhabit coniferous alpine and boreal forests.

(xi) Write a note on ocean thermal gradient.

(xii) Write any two factors which are responsible for modification of environment.

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) Compare Addison's disease and Cushing's disease.

(ii) Give the role of midbrain in humans.

(iii) What are neurotransmitters? How acetylcholine is different from other neurotransmitters?

(iv) Why Mendel uses Pisum sativum (Garden Pea) in his experiments?

(v) What is test cross? Write its significance.

(vi) What are multiple alleles?

(vii) What are restriction endonucleases? Give an example.

(viii) What is PCR? Write the role of taq polymerase.

(ix) What is gene therapy? What are its two methods?

(x) How the relationship of predator and prey is maintained?

(xi) Compare primary and secondary succession.

(xii) Differentiate between autecology and synecology.

Q4. (Marks 12) Write short answers to any SIX parts.

(i) Define neurulation. State events of neurulation and explain its significance.

(ii) What is morulla and blastula?

(iii) Interpret how many types of tRNA molecules are necessary for a living cell, if the genetic code is triplet code.

(iv) On the basis of position of centromere describe the four types of chromosomes.

(v) Differentiate between point mutation and chromosomal mutation.

(vi) Describe the symptoms and causes of down syndrome.

(vii) Compare mitosis with meiosis.

(viii) How does fossil record provide evidence of evolution?

(ix) What is genetic drift?

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5(a). (Marks 04) Explain the structure of nephron with the help of a diagram.

Q5(b). (Marks 04) What predation? Discuss its significance.

Q6(a). (Marks 04) How is support provided to those animals which lack a hard skeleton? Explain your answer with two examples.

Q6(b). (Marks 04) Describe the replication process of DNA in detail.

Q7(a). (Marks 04) Explain how reflex action prevent the body damage during emergency.

Q7(b). (Marks 04) Describe the importance of forest.

Q8(a). (Marks 04) Explain the process of birth in humans.

Q8(b). (Marks 04) Define diabetes and explain type-I in detail.

Q9(a). (Marks 04) Define growth correlation. Describe apical dominance in detail.

Q9(b). (Marks 04) Can the comparative anatomy and fossil record be discussed as evidence of evolution? Explain it.

GROUP – II

Biology (Objective) Group - II

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8462

> Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. The special excretory structures of terrestrial arthropods are called:

(A) Nephridia
(B) Protonephridia
(C) Malpighian tubules
(D) Nephrons

Q2. The plant tissues that lack secondary walls.

(A) Parenchyma
(B) Collenchyma
(C) Sclerenchyma
(D) Aerenchyma

Q3. How many pair of ribs are present in human rib cage?

(A) Ten pairs
(B) Twelve pairs
(C) Six pairs
(D) Twenty four pairs

Q4. The neurotransmitter that lie outside the central nervous system:

(A) Adrenaline
(B) Serotonin
(C) Dopamine
(D) Acetylcholine

Q5. The plant which is not a day neutral:

(A) Tomato
(B) Soya bean
(C) Cucumber
(D) Zea mays

Q6. The optimum temperature for maximum growth of plant is:

(A) 0 – 35 °C
(B) 25 – 30 °C
(C) 5 – 10 °C
(D) 10 – 15 °C

Q7. Grey equational cytoplasm give rise to:

(A) Notochord & neural tube
(B) Gut
(C) Muscle cells
(D) Larval epidermis

Q8. Each segment of DNA that is coiled around eight histone proteins:

(A) Karyotype
(B) Genotype
(C) Phenotype
(D) Nucleosome

Q9. In which syndrome the 21st pair of chromosome fails to segregate?

(A) Klinefelter's syndrome
(B) Turner's syndrome
(C) Down's syndrome
(D) Jacobs

Q10. In human cell the duration of G2 phase is:

(A) 10 hours
(B) 30 min
(C) 9 hours
(D) 4.5 hours

Q11. The chromosomes that carry its linked genes en bloc in the form of called:

(A) Linkage groups
(B) Linkage
(C) Recombinants
(D) Linked genes

Q12. The major goal of genetic engineer is to increase the efficiency of rubisco to introduce cycle:

(A) C2 cycle
(B) C4 cycle
(C) C3 cycle
(D) C6 cycle

Q13. The example of vestigial organ is:

(A) Nose
(B) Hairs
(C) Bones
(D) Vermiform appendix

Q14. The profession or job of any organism in their ecosystem called:

(A) Ecology
(B) Niche
(C) Autecology
(D) Synecology

Q15. The overall weather patterns are known as:

(A) Climate
(B) Weather
(C) Humidity
(D) Precipitation

Q16. The last biome seen before reaching the polar ice caps is:

(A) Taiga
(B) Boreal
(C) Arctic tundra
(D) Desert

Q17. Nuclear power station can last only for years:

(A) 10 years
(B) 20 years
(C) 30 years
(D) 40 years

Biology (Subjective) Group - II

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 8462

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) How are xerophytes adopted to survive in dry environment?

(ii) What is the structural differences between protonephridium and metanephridium?

(iii) Describe the procedure adopted to remove kidney stones.

(iv) Differentiate between heartwood and sapwood.

(v) What is sciatica?

(vi) What do you know about antagonistic arrangements of muscles?

(vii) What is the location and function of sertoli cells in male reproductive system of man?

(viii) Describe the process of generation of labour pain in human females.

(ix) What do you know about deserts of Pakistan?

(x) Describe profundal zone in fresh water lakes.

(xi) What is algal bloom?

(xii) Define soil and write its composition.

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) How nervous system of hydra differs from planaria? Give any two differences.

(ii) Which part of hind brain controls body movements and maintains position of body?

(iii) Write the effects of over secretion and under secretion of somatotropin hormone.

(iv) What are multiple alleles?

(v) What is meant by genic system?

(vi) What are sex limited traits?

(vii) What are restriction enzymes?

(viii) What is the significance of taq polymerase enzyme?

(ix) What is gene pharming?

(x) Differentiate between autecology and synecology.

(xi) What is ammonification?

(xii) What are decomposers? Give their significance in ecosystem.

Q4. (Marks 12) Write short answers to any SIX parts.

(i) Why morphogenetic determinants are essential in gene selection?

(ii) Discuss concept of differentiation.

(iii) Why CsCl is used for ultracentrifugation?

(iv) Discuss properties of DNA Polymerase-III

(v) Discuss findings of Archibald Garrod in certain genetic diseases.

(vi) Discuss importance of meiosis.

(vii) What are events of cell death?

(viii) What is artificial selection?

(ix) What is difference between endangered species and threatened species?

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5(a). (Marks 04) Write a detailed note on excretion in earthworm with the help of diagram.

Q5(b). (Marks 04) Explain principle stages of nitrogen cycle. Draw nitrogen cycle as well.

Q6(a). (Marks 04) Justify the division of joints on the basis of their structure.

Q6(b). (Marks 04) How DNA encodes protein structure? Discuss with special reference to control Dogma of molecular biology.

Q7(a). (Marks 04) How insulin depresses blood glucose levels. Also discuss diabetes mellitus.

Q7(b). (Marks 04) Describe the importance of forests.

Q8(a). (Marks 04) Describe the events of menstrual cycle and explain its hormonal regulation.

Q8(b). (Marks 04) Discuss Rh-blood group system in man.

Q9(a). (Marks 04) Write a note on differentiation in plants.

Q9(b). (Marks 04) Discuss any two factors affecting gene frequency.